



Trauma

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by HandTevy.
- Fluids and Medication titrated to maintain a SBP >70 + (age in years x 2) mmHg
- Hypotension: (SBP < 70+ 2x Age in years)

Signs & Symptoms:

- **M**assive Hemorrhage
- **A**irway
- **R**espirations (decompression)
- **C**irculation (IV, TXA)
- **H**ypothermia / Head injury
- **P**ain
- **A**ntibiotics
- **W**ound Care
- **S**plinting

Differential:

- Respiratory failure
- Foreign body airway obstruction
- Hypovolemia
- Trauma
- Tension pneumothorax
- Hypothermia
- Toxins or Overdose
- Hypoglycemia
- Acidosis
- Acute MI or PE

Clinical Management Options

P	P	P	P	P	P	• Control external hemorrhage and apply tourniquet(s) as necessary, including junctional tourniquets if needed and available. • Wound packing (junctional/extremity) with pressure dressing as appropriate and apply Quick Clot Combat Gauze if available • Elevate head of bed 30° unless cardiac arrest or step-offs noted in C-spine • BLS airway management • Place occlusive dressing/chest seal over open pneumothorax • Evaluate for spinal motion restriction – Do not transport laying on long spine board • Assess GCS score • Apply pelvic binder if appropriate • Keep patient supine and warm • Administer oxygen to targeted SpO₂ • Bandage/splint injuries as appropriate for patient condition • Declare Trauma Alert / Activation if appropriate for patient condition
L	L	L	L	L	L	• 3-lead ECG placement • ETCO₂ assessment • Acetaminophen (adult only) • Ibuprofen for musculoskeletal and/or joint pain/injury
1	2	3	4	5	6	• Acetaminophen (all routes) – IV Acetaminophen only if MAP ≥ 65 • Vascular access • Adult: Isotonic Crystalloid IV bolus 250 mL if patient shows signs of shock; Pediatric 20 mL/kg as needed • Tranexamic Acid (TXA)
						• Needle decompression of the chest as indicated • Advanced airway management • Pain management – Fentanyl and/or Ketamine as indicated. • If Adult Spinal Shock – Norepinephrine (Levophed) Infusion titrated to MAP ≥ 65 • Push-dose Epinephrine • Blood Product Transfusion and Calcium Chloride • Ultrasound for EFAST exam.
						• Simple Thoracostomy • Field Amputation as required • Escharotomy as indicated for ventilation compliance

Consult Online Medical Control As Needed



Trauma

GCS

Eyes Open	Best Verbal	Best Motor
4 – Eyes Open	5 – Oriented	6 – Obeys Commands
3 – To Voice	4 – Confused	5 – Localizes Pain
2 – To Pain	3 – Inappropriate	4 – Withdraws from Pain
1 – None	2 – Incomprehensible	3 – Pain-Flexion
	1 – None	2 – Pain-Extended
		1 – None

Pearls:

- Consider Chest Decompression with signs of shock and diminished/absent breath sounds. If the patient arrests, then immediately perform bilateral decompression.
- Minimize Scene time. If the patient meets Trauma Alert criteria, then interventions should be performed en route.
- Permissive hypotension (target fluid resuscitation to MAP 55-65) should be used in the absence of neurologic injury, pregnancy, hypertensive history, and age ≥ 45 years old.
- If neurologic injury – [Head Trauma COG](#)
- Peripheral neurovascular status should be documented on all extremity injuries and before and after splinting procedures.
- Document neuro status before and after extrication.
- Hip, knee, and elbow fractures/dislocations have a high incidence of neurovascular compromise.
- In hypotension, consider spinal shock or additional occult injury as the source.
- A decrease in mental status may be an early sign of shock.
- Avoid PO medications in patients who may require emergent surgical intervention